

fN=50;

fN =
50

$\omega_N = 2\pi \cdot f_N$

$\omega_N =$
314.1593

C=5012e-9;

C =
5.0120e-06

$X_c = 1/\omega_N/C$

$X_c =$
635.0955

IN=300

IN =
300

VN=110e3;

VN =
110000

$L_{Pet} = VN / (1.73205 \cdot IN \cdot \omega_N)$

$L_{Pet} =$
0.6738

$X_L = \omega_N \cdot L_{Pet}$

$X_L =$
211.6952

$L_{trafo} = 0.096/3$

$L_{trafo} =$
0.0320

$L_{line} = 0.0035/3$

$L_{line} =$
0.0012

$L_{add} = L_{trafo} \cdot 7/9$

$L_{add} =$
0.0249

$I_{pet} = VN / (1.73205 \cdot X_L)$

$I_{pet} =$
300

$f_{res} = (1/2\pi) \cdot \sqrt{1/(3 \cdot C \cdot (L_{Pet} + L_{trafo} + L_{add} + L_{line}))}$

$f_{res} =$
47.9764